

Market Brain Grid PRD

Market Brain Grid (MBG)

Product Requirements Document — Unified v2.0

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Multi-Agent Research & Trading Desk for IDX Equities + Crypto, on Hermes Agent

Status: Consolidated build baseline, merged from four independent draft PRDs (Grok v1.2, GPT/ChatGPT Final, GLM v1.1, Gemini/v5.0) plus the original brief. **Date:** 2026-09-08 **Not financial advice. Educational and research purposes only. Verify every number. Investing involves risk of loss.**

0. How this document was built

This PRD reconciles four independently generated drafts of the same brief. Rather than pick one draft, each section below was assembled from whichever source draft handled that topic best, with conflicts resolved explicitly (see §18, Open Decisions & Resolved Conflicts). Nothing here is invented — every requirement traces to at least one of the four source documents or to the original prompt.

Source lineage, for traceability: | Tag | Source draft | Strongest contribution | |—|—|—| | **[G]** | Grok v1.2 | Debate-mode catalog, Shapley credit assignment, concrete SKILL.md designs | | **[P]** | GPT/ChatGPT Final | Deterministic execution boundary, evidence ledger, storage/API contracts | | **[Z]** | GLM v1.1 (Z.ai) | Verified IDX microstructure rules, Hermes bot-profile mechanics, cost budget, MBG Lite | | **[V]** | Gemini v5.0 | Econometric proofing (DSR/PBO/CSCV/Hansen SPA), bandarmology, audit hash chain |

1. Executive Summary

Market Brain Grid (MBG) is a multi-agent research and **paper-first, human-gated** trading system covering Indonesian equities (IDX) and crypto, orchestrated by the **Hermes Agent** runtime. It produces a daily brief, on-demand deep-dive research, structured and auditable Trade Plans, and reproducible backtests — never a silent live order.

The architecture is deliberately a hybrid of five traceable lineages: - **TradingAgents** (arXiv:2412.20138) — specialist analyst fan-out + adversarial Bull/Bear debate + risk committee + portfolio manager, the only pattern in the research base with published backtest evidence. - **AutoHedge** — compact order/trade-plan schema and Director-style natural-language ticker discovery. - **Vibe-Trading** — skill-packaging discipline (SKILL.md + YAML frontmatter), data-quality-as-a-gate philosophy, walk-forward/Monte Carlo validation suite. - **Moss trade-bot skills** — natural-language strategy factory, bounded ($\pm 30\%$) parameter evolution with a locked core, backtest/live parity. - **The user-supplied Astra guide** — the SCAN → ANALYZE → RESEARCH → RISK → PLAN discipline, FACTS-vs-OPINION separation, and mandatory **AWAITING HUMAN REVIEW** status on every plan. **[G][Z]**

Core operating principle carried through every layer: the LLM proposes and explains; deterministic code validates data, calculates risk and position size, runs backtests, and gates execution. No path exists from natural language directly to an exchange order. **[P]**

Dimension	Decision
Runtime	Hermes Agent (Nous Research) — conversation/orchestration layer, bot profiles, skills, cron
Compute	Python 3.11+ deterministic services (data, risk, backtest, execution) running under Hermes's supervision, not as free-form LLM reasoning [P][Z]
Markets	Indonesia IDX/BEI equities + crypto spot (v1); crypto derivatives explicitly gated to a later, compliance-reviewed phase [Z][P] vs [V] — see §18
State/DB	Supabase (Postgres + Realtime + Storage + pgvector); Firebase optional, client/event layer only [P][Z][V]
Execution policy	Paper/Research/Shadow by default. Guarded Live only after explicit phase-gated human approval + kill-switch. [G][P][Z][V]
Credit assignment	Online Shapley-value agent/mode weighting once sufficient closed trades exist [G]
Strategy validation	Walk-forward, Monte Carlo/bootstrap, and (optional, P2+) econometric proofing: Deflated Sharpe Ratio, PBO/CSCV, Hansen SPA [Z][V]

2. Product Vision, Goals, Non-Goals

2.1 Vision

Give one Indonesian retail investor the research depth of an institutional desk without ever letting a language model move money on its own: a team of specialist AI analysts, in permanent adversarial tension with each other, that documents every argument, is graded against its own outcomes, and stops and says **NO TRADE** when evidence or data quality is insufficient. **[Z][P]**

2.2 Goals (v1)

- Daily brief (IDX + crypto) delivered before the IDX pre-open. **[Z]**
- On-demand deep-dive research for a single ticker/pair, following SCAN→ANALYZE→RESEARCH→RISK→PLAN. **[G][Z]**
- Structured, auditable Trade Plans: entry, stop, targets, position size (with visible arithmetic), R:R, invalidation, confidence, expiry — always ending **STATUS: AWAITING HUMAN REVIEW**. **[G][Z][P]**
- Multi-mode adversarial/collaborative debate with a Research Manager synthesis step. **[G][Z][V]**
- Reproducible backtesting with point-in-time data, realistic costs, and leakage checks. **[P][Z][V]**
- Paper trading portfolio with realistic settlement simulation (IDX T+2). **[Z][P]**
- Online Shapley credit assignment to learn which agents/modes actually contribute to outcomes. **[G]**
- Deployable laptop-first, cloud-state-plane-second (Supabase); Firebase optional. **[Z][P]**
- Strict safety: paper-first, human gate for any live action, kill-switch, full audit trail. **[G][P][Z][V]**

2.3 Non-Goals (v1)

- Fully unsupervised live capital deployment, in any market. **[G][P][Z]**
- High-frequency / sub-second market making. **[G][P]**
- Real-money order execution on IDX brokers (no Indonesian retail broker publishes a public trading API as of this writing — see §7.3). **[Z]**
- Crypto derivatives / leveraged perpetuals trading (only spot, on OJK-whitelisted venues) until a dedicated compliance and risk review is complete. **[Z][P]** — this explicitly supersedes v5's perpetuals-first design; see §18.
- Multi-user SaaS, tenancy, or payments. **[Z]**
- Tax, accounting, or regulatory filing automation. **[G]**
- Automatic, unreviewed production parameter mutation. **[Z]**

2.4 Design Philosophy

1. **Advise before act** — trust is earned through a measured, honestly-scored paper-trading record before any live pathway is even designed. **[Z]**
2. **Structure over prose** — every agent emits a machine-readable, schema-validated JSON artifact; prose is reserved for the debate transcript and the human-facing brief. **[Z][P]**
3. **Honest measurement** — every recommendation is logged with its timestamp, the data available at decision time, and its eventual outcome, so the reflection memory and backtester compute genuine point-in-time alpha, not after-the-fact narrative. **[Z][G]**
4. **LLM output is an untrusted input** to the risk engine, order gateway, policy engine, and reconciliation service, which remain the system of record for anything involving money. **[P]**
5. **Never invent a number.** If data is missing, the output says **NOT PROVIDED – verify manually** instead of guessing. **[G][Z]**

3. Universal Safety Rules (Mandatory in Every Skill / Every Agent Prompt)

These are non-negotiable and must appear in every MBG skill prompt and every research/plan output, verbatim in spirit: **[G][Z]**

1. Never invent a price, number, date, headline, or metric. Missing data → **NOT PROVIDED – verify manually**.
 2. Keep **FACTS** (with source + timestamp where available) strictly separate from **OPINION**.
 3. No profit promises, no guaranteed returns, no buy/sell orders in natural language. Language stays research-oriented.
 4. Every Trade Plan ends with **STATUS: AWAITING HUMAN REVIEW** — nothing executes until a human approves a specific plan ID.
 5. Position sizing must show its arithmetic: $(\text{account} \times \text{risk\%}) \div (\text{entry} - \text{stop}) = \text{size}$.
 6. If Risk/Reward is under ~1.5–2:1 at the proposed entry (asset-class dependent, see §11), the plan must say explicitly that it is not worth taking at this price.
 7. Paper mode is default. Live mode requires explicit human confirmation per-plan and is subject to kill-switch and mandate limits.
 8. Full audit log of every agent decision, tool call, data fetch, and order proposal — hash-chained for tamper evidence. **[V]**
 9. A missing prerequisite (stale data, unresolved instrument identity, failed data-quality gate) returns a typed **BLOCKED** status, never a fabricated answer. **[P]**
 10. Every published output carries a standing disclaimer: educational content, not financial advice; the system can be wrong; the user bears full responsibility for any action taken. **[Z]**
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4. System Architecture

MBG uses a **three-plane hybrid architecture** that separates conversation, computation, and state — this is the one structural decision on which all four source drafts converge. [Z][P]



[Z, Figure 1, adapted]

4.1 Why this split (not “just run it all as Hermes skills”)

Serverless platforms cannot host a 24/7 Python agent process: Supabase Edge Functions cap at 150–400s wall-clock; Firebase Gen-2 Cloud Functions cap at 540–3,600s. [Z] A laptop (or VPS/Docker) worker with a restart policy and a database heartbeat handles this natively. Hermes is therefore the **conversation and workflow-routing layer** — it owns bot personas, skill discovery, cron, and the chat surface — while the actual math (risk, sizing, backtests, order validation) lives in deterministic Python services that Hermes’s skills call into. [P] This is the resolution to the Grok/GLM-vs-GPT tension noted in §18.1: skills are still the right unit for *repeatable workflows*, but a skill that computes position size must call a pinned, testable Python function, not re-derive arithmetic from an LLM each time.

4.2 Component responsibilities

Layer	Responsibility	Source
Orchestration (Hermes Gateway)	Bot profiles, SKILL.md skill loading, cron routines (daily brief, EOD), Telegram/CLI/dashboard delivery, group-chat handoffs	[Z][G]
Data Steward	OHLCV, fundamentals, news, on-chain; point-in-time; multi-source fallback chain; identity/unit/freshness validation	[P][Z]
Analysis	Technical, Fundamental, Sentiment/News, Macro specialists, market-context split (IDX vs crypto)	[G][P][Z][V]
Debate & Research	Bull/Bear (+ optional collaborative/gated/regime-conditional/contest modes), Research Manager synthesis	[G][Z][V]
Trading & Risk	Trader → Risk Committee (aggressive/neutral/conservative) → Portfolio Manager; conviction gating	[Z][P][V]
Credit & Memory	Shapley credit assigner, reflection memory, outcome linking	[G][Z]
Econometric Proofer (optional, P2+)	DSR, PBO/CSCV, Hansen SPA independent mathematical veto on any strategy before it enters the daily loop	[V]
Backtest & Lab	Strategy simulation, walk-forward, Monte Carlo/bootstrap, stress tests	[P][Z][V]
Execution	Paper broker (default); shadow mode; guarded-live adapters (Phase 3+); kill-switch	[G][P][Z][V]

4.3 Technology stack

Layer	Choice	Rationale
Bot runtime	Hermes Agent (Python 3.11+, MIT)	Native SKILL.md skills, bot profiles, cron, Telegram gateway, multi-bot messaging [Z]
Agent graph	LangGraph ≥ 0.4 (or equivalent state machine inside Python services)	Proven multi-node topology, checkpoint/resume, conditional debate loops [Z]
LLM access	Provider-agnostic adapter (Anthropic/OpenAI/Google/DeepSeek/local) with per-role model tiers (fast vs deep)	Cost control; vendor swap is config, not code [Z][P]
IDX data	yfinance (.JK) + idx.co.id scraper (curl_cffi, Cloudflare-aware) + Kontan/Bisnis RSS	Free baseline coverage; licensed feed as a documented future upgrade [Z][P]

Crypto data	ccxt (Indodax, Tokocrypto — OJK-whitelisted) + CoinGecko/CoinMarketCap free tiers	Both verified working live via ccxt during research [Z]
Backtesting	vectorbt (vectorized signal replay) + freqtrade (crypto strategy workbench)	10-year, 20-ticker daily replay in seconds [Z]
State	Supabase Postgres 15 + Realtime + Storage + pgvector	Relational fit for OHLCV/ledgers; generous free tier; RLS [P][Z]
Dashboard	Next.js/React on Vercel free tier, subscribed to Supabase Realtime	Read-mostly operator console; one write path (the worker) [P][Z]
Packaging	Docker Compose (worker) + SKILL.md bundles (bots)	One-command laptop bring-up; skills portable to OpenClaw/MOSS/Claude Code [Z]

Important architectural rule, carried from [P]: never embed exchange secrets or direct order methods inside analyst/researcher skills. Only a hardened Execution Gateway service may construct an order request, and every request passes deterministic pre-trade checks before it can be sent.

5. Bot / Agent Roster

MBG expresses each reasoning role as a **named Hermes bot profile** (persona in `SOUL.md`, pinned model tier, skill allowlist, own memory namespace) backed by a **deterministic tool/skill** for anything involving numbers. [Z][G][P]

5.1 Analysis & decision bots

Bot	Model tier	Trigger	Inputs	Structured output	Hard boundary
<code>mbg-data-steward</code>	fast	scheduled (05:45 WIB)	vendor fallback chain	<code>NormalizedCandleBuffer</code> / data-quality row	May not form a trade thesis; strict NaN fail-closed [Z][V]
<code>mbg-technical</code>	fast	pipeline run	OHLCV, verified price snapshot	<code>technical_report</code> (trend, momentum, S/R, volatility)	No order creation [Z][P]
<code>mbg-fundamental</code>	fast	pipeline run (IDX only, skipped for crypto)	financial statements, ratios	<code>fundamentals_report</code>	Facts must be sourced and dated [Z][P]
<code>mbg-idx-specialist</code>	fast	IDX pipeline run	broker summary, foreign flow, ARA/ARB bounds	<code>IDXMicrostructureReport</code> (accumulation status, broker concentration %, tick size)	No order creation [V]
<code>mbg-crypto-specialist</code>	fast	crypto pipeline run	funding rate, OI, liquidation clusters, order-book depth	<code>CryptoDerivativesReport</code> / <code>CryptoSpotReport</code>	Derivatives fields populated only if Phase 4 gate is open [V][Z]
<code>mbg-sentiment</code>	fast	pipeline run	Stockbit/X buzz proxies, funding/flow	<code>sentiment_report</code> (band, score, confidence)	Never infer a price from a headline alone [Z][P]
<code>mbg-news</code>	fast	pipeline run	Kontan/Bisnis RSS, IDX disclosures, macro calendar	<code>news_report</code> , rumor vs confirmed	— [Z][G]
<code>mbg-macro</code>	fast	pipeline run	rates, FX (USD/IDR), commodities, policy calendar	<code>macro_report</code>	No instrument-specific execution [P]
<code>mbg-quant</code>	fast	pipeline run	all analyst facts	<code>AlphaSignalMatrix</code> — 5-pillar (Trend/Momentum/Reversion/Volume/Volatility) + Smart Money Concepts (FVG, order blocks, liquidity sweeps)	Must distinguish computed values from sourced facts [V][P]
<code>mbg-regime-detector</code>	fast	pipeline run	price, volume, macro proxies	<code>RegimeLabel</code> (trend × volatility) + confidence	Consumed by debate-orchestrator and Shapley regime multipliers [G]
<code>mbg-bull</code>	fast	debate round	analyst reports + debate history	<code>bull_argument</code> — strongest defensible case, every claim cited	Cannot approve its own thesis [Z][G][V]
<code>mbg-bear</code>	fast	debate round	analyst reports + debate history	<code>bear_argument</code> — must rebut Bull's strongest point first	Cannot approve its own thesis [Z][G][V]
<code>mbg-research-</code>	deep	after N debate	full transcript	<code>investment_plan</code> — 5-tier verdict	Cannot bypass

manager		rounds		(Buy/Overweight/Hold/Underweight/Sell), conviction 0–1, key risks; Hold when balanced	the risk gate [Z][G][P]
mbg-trader	fast	verdict received	investment_plan + technical_report	trader_proposal — entry, stop, targets, absolute prices	No order creation; proposal only [Z][V]
mbg-risk-aggressive / -neutral / -conservative	fast	risk debate	trader proposal + all reports	risk_opinion (approve/reject + reasons)	Hard veto authority collectively [Z][P]
mbg-conviction-gater	fast (deterministic)	after debate	conviction score, portfolio constraints	GatedPlan or HOLD with reason	Threshold θ default 0.35–0.5 [G]
mbg-portfolio-manager	deep	risk debate closed	everything + memory past_context	portfolio_decision — final, publishable	Never directly calls exchange APIs [Z][P][G]
mbg-econometric-proofer (P2+, optional)	deterministic	before any strategy enters the daily loop	trade/strategy history	ProofingCertificate (DSR $\geq$ 0.95, PBO $\leq$ 0.15, Hansen SPA p-value)	Independent hard veto power, cannot be overridden by conviction score [V]

5.2 Infrastructure / non-analytical bots [Z][G][P][V]

Bot	Role	Notes
mbg-risk-guard	Deterministic pre-trade validation service (chat-facing profile for reporting only)	Enforces §11 rulebook on every plan and every trigger; owns the kill-switch
mbg-shapley-credit-assigner	Scheduled (weekly) Shapley-value computation over closed trades	See §6
mbg-strategist-memory	Decision log, outcome linking, reflection lessons, watchlist-level sector memory	Critical path for credit assignment — without outcome linkage, Shapley cannot run
mbg-execution-gateway	Deterministic order-ticket construction, prechecks, reconciliation	No LLM reasoning inside the critical path; never accepts free text as order input
mbg-reporter	Assembles daily brief and EOD scorecard from structured outputs only	Formatter, not an opinion source — degraded sections are labeled, never silently omitted
mbg-strategy-evolver (P1+)	Bounded parameter evolution ($\pm 30\%$ drift, core logic locked)	Rejects any change without out-of-sample improvement; cannot mutate production config automatically

5.3 Hermes mechanics mapping [Z]

MBG concept	Hermes mechanism
Bot roster	Hermes Bot Mode profiles under <code>~/.hermes/profiles/</code> , each with <code>SOUL.md</code> persona, model pin, skill allowlist
Skill catalog	<code>~/.hermes/skills/mbg/*/SKILL.md</code> packages, each with optional <code>scripts/</code> (see §5.4)
Daily 07:00 / EOD 16:15 WIB jobs	<code>hermes cron</code> routines attached to the reporter/steward profiles (WIB timezone, no-agent mode invoking scripts directly)
Analyst → debate → trader handoffs	Group chat with <code>@mentions</code> for visibility; actual data handoff via shared state store, not chat text (prevents “telephone game” degradation)
Bot-to-bot requests (e.g. trader asks steward for fresh bars)	<code>message_agent</code> DMs with typed failure reasons
Telegram commands	Gateway long-polling connection; skills register <code>/brief</code> , <code>/analyze</code> , <code>/plan</code> , <code>/backtest</code> , <code>/portfolio</code> , <code>/risk</code> , <code>/status</code> , <code>/kill</code>
Future cross-machine deployment	<code>hermes peer</code> bot-to-bot channel

5.4 Skill directory layout [G, extended with Z’s bot-profile split]

```

~/hermes/skills/mbg/
├── mbg-research-desk/          # Astra-style one-shot SCAN->ANALYZE->RESEARCH->RISK->PLAN
├── mbg-lite/                  # zero-infrastructure onboarding tier (see §14.4)
├── debate-orchestrator/       # selects mode, runs state machine, logs
├── debate-adversarial-bull-bear/
├── debate-safe-collaborative/
├── debate-consensus-gated/
├── debate-regime-conditional/
├── debate-contest/
├── research-manager/
├── regime-detector/
├── conviction-gater/
├── risk-manager/
├── shapley-credit-assigner/
│   └── scripts/shapley_exact.py
├── econometric-proofer/      # P2+, optional
│   └── scripts/dsr.py, cscv.py, hansen_spa.py
├── strategist-memory/
├── data-fetcher/
├── technical-analyst/
├── fundamental-analyst/
├── idx-specialist/
├── crypto-specialist/
├── sentiment-news-analyst/
├── backtest-lab/
└── execution-gateway/       # locked down: no bare LLM calls in this skill's critical path

~/hermes/profiles/           # bot personas (SOUL.md) referencing the skills above

```

Each skill directory contains a `SKILL.md` (frontmatter + procedure) and, where it touches numbers, a `scripts/` folder of pinned, unit-tested Python. **[G][Z]**

6. Debate Strategies & Agent Credit Assignment

6.1 Debate modes **[G]**

Skill / Mode	Description	Prefer when
<code>debate-adversarial-bull-bear</code>	1-2 round Bull vs Bear + Research Manager synthesis	Ambiguous directional thesis (this is also the TradingAgents-evidenced default — [Z][P][V] all converge here)
<code>debate-safe-collaborative</code>	Single-round Trust/Skeptic/Leader refinement	Report quality, multi-horizon research
<code>debate-consensus-gated</code>	Debate then conviction/consensus threshold θ	High-noise conditions (crypto), risk-first
<code>debate-regime-conditional</code>	Competitive vs collaborative selection by detected regime	Multi-regime IDX + crypto
<code>debate-contest</code>	Parallel paths scored after outcomes; re-allocate	Long-running continuous improvement
<code>debate-orchestrator</code>	Meta-skill: selects mode, runs state machine, persists transcript	Always the entry point

Pitfall to guard against explicitly: do not default to high round counts (cost); always store `transcript_id` for later Shapley joins. **[G]**

6.2 Shapley value agent credit assignment **[G]**

Credit assignment answers: *which agent or coalition actually contributed to profitable outcomes?* The Shapley value is the unique fair division satisfying efficiency, symmetry, dummy, and additivity:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! \cdot (|N| - |S| - 1)!}{|N|!} \cdot [v(S \cup \{i\}) - v(S)]$$

For $N \leq 4$ specialists, exact enumeration of $2^N - 1$ coalitions is practical. $v(S)$ is defined from realized R-multiples or risk-adjusted contribution over an exponentially-weighted window — **never from LLM self-confidence**.

Procedure: load exponential-decay performance histories → enumerate coalitions → compute exact ϕ_i → normalize to weights ω_i → Bayesian-mix with a uniform prior during cold-start → optionally scale by regime multipliers from `mbg-regime-detector` → persist `CreditReport` + causal trace for audit → recompute on a schedule (e.g. weekly) once sufficient closed trades exist.

Verification: $\sum \omega \approx 1$; dummy (non-contributing) agents' weights trend near zero; report is human-readable. **[G]**

7. Market Context and Problem Statement

7.1 Why IDX specifically needs this

Indonesian retail investors face a structural information gap: the IDX lists 900+ equities, but quality research coverage concentrates on the 50–100 most liquid names. Sell-side reports are scarce and Stockbit community discussion is noisy. None of the four reference open-source trading-agent projects (TradingAgents, AutoHedge, Vibe-Trading, Moss) ships an IDX data adapter — this gap is original work MBG must build. [Z]

7.2 IDX market structure (verified rules) [Z]

Rule	Verified value	System impact
Session I	09:00–12:00 WIB (Fri 09:00–11:30)	Daily brief must land before 08:30 WIB
Session II (regular)	13:30–15:49:59 WIB (Fri 14:00–15:49:59)	Intraday trigger window
Pre-closing / closing	15:50–15:59:59; closing match ~16:00–16:09:30	EOD snapshot scheduled 16:15 WIB; triggers never fire in the final 10 minutes of regular trading
Tick fractions	Rp1 / Rp2 / Rp5 / Rp10 / Rp25 by price band	Limit-price rounding on every plan card
Auto rejection (ARA/ARB)	ARA ~35%/25%/20% by price tier; ARB 15% (Main Board)	Order-price sanity check before user confirmation; forbids unrealistic backtest fills at a locked-limit price
Board lot	1 lot = 100 shares	Position size always expressed in lots
Settlement	T+2	Paper broker cash-availability ledger; buying power respects the same rule
Illiquid stocks	Full call auction since Mar 2024	Liquidity filter excludes illiquid names from the intraday engine

7.3 The broker API reality [Z]

No Indonesian retail securities firm (Stockbit Sekuritas, Mirae Asset Sekuritas, IPOT, or others) currently publishes an official retail trading API for IDX equities. Unofficial scrapers exist but violate terms of service and risk account suspension. **This reshapes the product for v1: IDX “execution” is a one-tap confirmation flow that hands a validated, tick-correct order ticket to the human user; nothing is transmitted automatically.**

7.4 IDX research dimensions [P][V]

- Price/volume trend, relative strength, volatility, support/resistance.
- Financial statement trend, margins, cash/debt, valuation, earnings revisions, corporate actions.
- **Bandarmology** — Top-3 broker net accumulation share. If Top-3 buyer volume > 65% of session volume while price consolidates within ~1.5× ATR, flag an Accumulation signal. [V]
- Foreign/domestic flow proxies where reliable and permitted.
- Commodity and FX sensitivity for relevant sectors.
- Government/regulatory/event catalysts; event-timing risk gate blocks new entries into earnings/corporate-action windows.
- Liquidity, gap risk, price-band behavior for the proposed position size.

7.5 Crypto regulatory shift (verified) [Z]

Oversight of crypto assets in Indonesia moved from Bappebti to OJK (with Bank Indonesia responsible for payment-related aspects) effective 10 January 2025, transition completed within a 24-month window under the P2SK law. OJK published a whitelist of licensed trading platforms in December 2025, including Indodax, Tokocrypto, Pintu, Reku, Triv, and Pluang. **Binance global is not licensed to serve Indonesian residents.** Trades on local exchanges carry a 0.1% final income tax plus 0.11% VAT under PMK 68/2022.

MBG therefore treats the OJK whitelist as its v1 execution universe. Market data may be pulled from any source via the exchange-agnostic `ccxt` layer, but any future order routing targets whitelisted venues only, and derivatives/leveraged perpetuals are out of scope until a dedicated compliance review — see §18.2 for why this directly overrides the v5 draft’s Hyperliquid/OKX perpetuals design.

7.6 Crypto research dimensions [P][V]

- Price/volume/momentum and volatility regimes.
- Order-book depth, spread, imbalance, liquidity decay.
- Exchange flows, on-chain activity, token-specific events where available.
- *(Derivatives-only, Phase 4+ if approved)* funding rate drag, open-interest divergence, liquidation-cluster heatmaps; minimum liquidation buffer must exceed 3.5× daily ATR before any derivatives feature activates.
- BTC/ETH dominance / benchmark-relative behavior.
- 24/7 event risk, exchange-outage risk, stablecoin/liquidity risk.

7.7 Cross-market macro layer [P]

A macro layer contextualizes but does not dictate individual trades: USD/IDR, global rates, commodities relevant to Indonesia, global equity risk sentiment, Bitcoin risk regime, major central-bank/domestic policy events. Macro scores stay explicitly separate from instrument-specific evidence.

8. Data Architecture and Provenance

8.1 Canonical data flow [P]

Providers -> Raw Capture -> Normalizer -> Quality Gate -> Feature Store -> Evidence Ledger -> Agent Context -> Decision Artifacts

8.2 Evidence object schema [P]

```
{
  "evidence_id": "ev ...",
  "instrument_id": "IDX:BBCA",
  "kind": "price|fundamental|news|macro|onchain|derived",
  "observed_at": "2026-09-08T09:00:00+07:00",
  "published_at": "...",
  "source": "provider_name",
  "value": 123.45,
  "unit": "IDR",
  "price_caliber": "raw|adjusted|unknown",
  "payload_hash": "sha256...",
  "quality": "PASS|WARN|FAIL"
}
```

8.3 Mandatory data-quality gates [P]

Gate	Reject / warn condition
Identity	Ambiguous ticker, wrong market or quote currency
Freshness	Data older than the policy threshold for the requested timeframe
Completeness	Missing bars, missing corporate-action records, missing critical fields
Units	Volume or contract size cannot be identified with confidence
Caliber	Adjusted/raw price treatment mixed without explicit reconciliation
Time	Published/announcement time unavailable when event timing matters
Source integrity	Provider returns HTML/error/empty payload where numeric data is expected
Cross-source discrepancy	Primary vs secondary values differ beyond configured tolerance
Lookahead	Evidence was not public by the simulated decision timestamp

8.4 Provider abstraction [P][Z]

Adapter	Responsibilities	Sources
IDX Market Data	OHLCV, corporate actions, calendar, instrument metadata	yfinance (.JK , verified back to 2004), idx.co.id (curl_cffi), licensed feed as future upgrade
Crypto Market Data	Ticker, OHLCV, trades, order book	Native exchange APIs + ccxt (Indodax, Tokocrypto verified live)
Fundamental Data	Financial statements, corporate events	Provider adapters, company filings
News	Timestamped news/event feed	Kontan/Bisnis RSS, IDX disclosures scrape
Sentiment	Social/public chatter	Careful source labeling; never treated as verified fact
Macro	Rates, FX, commodities, economic calendar	Authoritative economic/market sources

8.5 Data snapshot & fingerprinting [P][Z]

Every backtest and high-impact recommendation stores its data snapshot version, parameter set, code version, provider versions, and rule configuration together — essential for debugging and for proving a silently-changed provider didn't invalidate a backtest result.

9. End-to-End Workflow & Decision Lifecycle

9.1 Daily desk master workflow [P][Z]

SCHEDULE -> DATA HEALTH -> SCAN -> ANALYZE -> RESEARCH -> DEBATE -> RISK -> PORTFOLIO DECISION -> PLAN -> ALERT -> PAPER/SHADOW -> JOURNAL

9.2 Astra 5-step workflow, upgraded [P]

Stage	Original idea retained	System upgrade
SCAN	Find 3-5 candidates; rank clarity, not return	Multi-market screener + data-quality score + liquidity filters + portfolio context; zero LLM tokens in the ranking pass [Z, F-10]
ANALYZE	Trend, momentum, S/R, price action	Deterministic technical features + chart narrative; facts kept visibly separate from interpretation
RESEARCH	Financials, earnings, news, catalysts	Source graph + event-time knowledge + IDX-vs-crypto research templates
RISK	Stop, size, R:R, invalidation	Portfolio-aware risk budget + stress tests + correlation + liquidity + hard policy veto
PLAN	One structured plan awaiting review	Proposal object with evidence IDs, expiry, scenario tree, execution constraints, status machine

9.3 Candidate state machine [P]

DISCOVERED -> DATA_VALIDATED -> ANALYZED -> DEBATED -> RISK_CHECKED -> PROPOSED -> PAPER
-> APPROVED -> EXECUTED -> MONITORED -> CLOSED -> REVIEWED

Any stage -> BLOCKED / REJECTED / EXPIRED

Any live position -> RISK_BREACH -> EMERGENCY_ACTION -> RECONCILED

9.4 Recommendation statuses [P]

Status	Meaning
NO SETUP	No sufficiently clear opportunity
WATCH	Interesting but missing confirmation or price condition
PAPER CANDIDATE	Passes research + deterministic risk in simulation
LIVE PROPOSAL	Ready for review under live policy; not an order
APPROVED	User explicitly approves a specific proposal ID and constraints
REJECTED	Failed policy, evidence, or review
EXPIRED	Conditions changed or data is too old; must be regenerated

9.5 Operating modes [P]

Mode	Behavior	Execution
RESEARCH	Collect/analyze/report; no simulated positions required	NONE
PAPER	Simulated portfolio; real-time or delayed data	SIMULATED
SHADOW	Observe a live account, report what MBG <i>would</i> have done	NONE
GUARDED LIVE	Live orders only after explicit approval + risk-gateway checks	ALLOWLISTED, small notional
EMERGENCY STOP	Cancel/reject new risk-taking orders; reconcile state	NONE / CLOSE-ONLY if separately authorized

9.6 Daily operating cadence (WIB) [Z]

Time	Job
05:30	Opportunity scanner (non-LLM screen, top-5 candidates appended to brief)
05:45	Data health refresh + integrity checks

06:15	Per-ticker deep analysis of watchlist (staggered for rate limits)
07:00	Consolidated Telegram/dashboard daily brief
08:15	IDX-specific proofed trade-plan delivery (if econometric proofer enabled) [V]
Intraday	5-minute IDX bars during both sessions; 15-minute crypto cadence; lightweight trigger conditions only (never the full pipeline)
16:15	EOD snapshot: scores yesterday's plans, updates paper portfolio, queues memory reflections
Crypto 24/7	Event-driven alerts + scheduled 4h/8h/24h regime summaries; 8-hour funding-cycle alerts if derivatives phase is active [V]
Weekly	Strategy diagnostics, parameter review, Shapley recompute, trade-journal behavior review
Monthly	Risk-policy review, data-source audit, model cost/quality review

10. Backtesting and Validation Framework

10.1 Design principles [P][Z]

- Backtests use the **same strategy interface** as paper/live execution (bit-exact parity, adopted from Moss).
- Signal time and fill time are distinct: a signal from bar T cannot fill at a price unavailable until after the decision point.
- Fees, slippage, spread, market impact, and funding/financing are configurable and always reported.
- Corporate-action and price-adjustment policy is explicit for equities.
- Execution constraints are venue-specific (sessions, lot size, leverage, short availability).
- Every result carries a data-quality verdict and a leakage check; a run without its benchmark row is invalid and the UI marks it as such.

10.2 Backtest types [P]

Type	Use
Single-strategy	Validate one strategy on one instrument/universe
Portfolio backtest	Multi-asset allocation, shared capital and risk budget
Walk-forward	Train on a rolling window, score on a later unseen window (default 12mo train / 3mo test)
Monte Carlo / bootstrap	Reshuffle trade order, bootstrap returns; confidence bands on max drawdown
Stress test	Shock spread, slippage, volatility, gaps, correlated drawdowns
Event replay	Re-run around specific news/catalyst periods
Shadow comparison	Compare proposal against naive baseline or benchmark
Evolution backtest	Bounded parameter search with fixed personality/risk constraints

10.3 Minimum metric set [P][Z]

Category	Metrics
Return	Cumulative return, CAGR, benchmark-relative return
Risk	Max drawdown, downside deviation, VaR/ES, volatility
Trading	Win rate, payoff ratio, expectancy, turnover, avg hold time
Efficiency	Sharpe, Sortino, Calmar (methodology always disclosed)
Execution	Avg slippage, spread cost, fee cost, fill ratio, rejected orders
Robustness	Out-of-sample score, parameter sensitivity, regime breakdown, bootstrap confidence

10.4 Anti-leakage checklist [P]

- Use `published_at`, not `period_end`, for news/fundamental availability where applicable.
- Never fill at the same close used to calculate the signal, unless the strategy explicitly models auction/close execution.
- Separate universe selection from future constituents to avoid survivorship bias.
- Use historically valid corporate-action data; never mix adjusted prices with raw execution assumptions.
- Prevent future funding rates, future index membership, or post-event labels from entering prior bars.
- Use the lot/contract/fee rules that would actually have been known at the time.

10.5 Optional: econometric strategy proofing (P2+) [V]

Before any strategy variant is promoted into the daily loop, an independent, non-LLM referee can (configurable, off by default until validated on your own trade history) apply: - **Deflated Sharpe Ratio (DSR ≥ 0.95)** — penalizes return by the number of trials logged in a `TrialLedger`, adjusted for skewness/kurtosis, guarding against data-snooping across many backtested variants. - **Combinatorially Symmetric Cross-Validation (PBO ≤ 0.15)** — slices history into blocks (e.g. 16 blocks → C(16,8)=12,870 combinatorial splits) to estimate the Probability of Backtest Overfitting. - **Hansen SPA test** — bootstrap test of whether the “best” strategy actually beats a benchmark once multiple-testing is accounted for.

This module is powerful but heavy; it is explicitly optional and gated behind Phase 2+ (see §16) rather than a v1 requirement, because it assumes a strategy library large enough to make overfitting a real risk — premature for a single-user, single-strategy launch.

11. Risk Management and Execution Safety

Safety rule, carried through the entire stack: LLM outputs are untrusted inputs. The risk engine, order gateway, policy engine, and reconciliation service are the system of record for money-moving state. [P]

11.1 Risk hierarchy [P]

Portfolio limits -> Strategy limits -> Instrument limits -> Position sizing -> Order limits -> Exchange/broker constraints

11.2 Hard limits (defaults, all configurable in `config/risk.yaml` except where noted) [Z][V][P]

Control	Default posture	Configurable?
Max risk per trade	1.25-2% of equity (moderate profile; see worked example below)	YES
Max single-name exposure	20% of equity	YES
Max sector exposure (IDX)	35% of equity	YES
Daily portfolio stop	−6% from prior close → arms soft kill, new longs blocked until next session	YES
Circuit-breaker daily drawdown	> 2.5% cumulative NAV loss in 24h → all open orders canceled, trading halted 24h	YES
Max drawdown kill	−10% from equity peak → arms hard kill until manual release	YES
Mandatory stop-loss	Every plan; cards without stops are invalid by schema	NO
Min R:R ratio	1.5-2:1 depending on asset class; below this the plan must say it's not worth taking	YES
Max open plans	8 concurrent	YES
Crypto leverage	Spot only in v1; 1x default until derivatives are separately approved	YES, gated
Liquidity cap	Position/order size limited by estimated executable liquidity	YES
Stale-data block	No live order when required price/account data is stale	YES
Kill switch	Reject new orders, surface emergency state	NO — mandatory, cannot be removed
Manual approval	Required before first live deployment and configurable after	Cannot be removed from high-risk operations

11.3 Position sizing formula (shown on every plan card) [Z][P]

```
risk_budget = portfolio_equity * max_risk_pct
risk_per_unit = abs(entry_price - stop_price)
raw_qty = risk_budget / risk_per_unit
final_qty = floor_to_venue_constraints(raw_qty)      # lots for IDX, min increment for crypto
notional = final_qty * worst_case_entry_price
portfolio_after = current_exposure + proposed_exposure
```

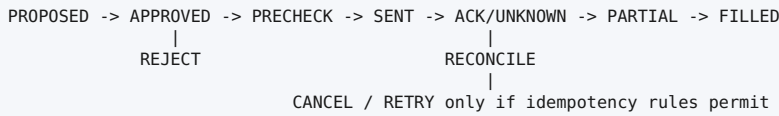
The system must use the *worse* of quote price and limit price when sizing a limit order, and must reject ambiguous/unparsable prices instead of silently defaulting. [P]

Worked example [Z]: equity Rp 100,000,000; BBCA plan entry Rp 9,800 / stop Rp 9,500 → risk per share Rp 300. At 2% risk, budget = Rp 2,000,000 → formula yields 666 shares → floored to 6 lots (600 shares, Rp 5,880,000 notional, ~5.9% of equity, actual risk Rp 1,800,000 = 1.8%). The plan card shows both the formula result and the floored execution size.

11.4 Scenario tests before any live proposal [P]

Normal case (expected spread/slippage) · Worse fill (spread + slippage multiplier) · Gap case (stop filled materially beyond stop price) · Liquidity case (only a fraction of desired quantity fillable) · Correlated shock (portfolio correlation rises while the trade moves against it) · Venue outage (ambiguous submission; must reconcile before retry).

11.5 Execution state machine [P]



11.6 Emergency behavior [P][V]

Emergency stop is a **system state**, not a chat instruction. When active, all new risk-taking orders are rejected; the system surfaces open positions, stale account state, exchange health, and outstanding order uncertainty. Any close-only or hedge action has a separate, deterministic emergency policy. Kill levels: **soft** (pause new analyses), **hard** (pause analyses and triggers), **panic** (additionally flatten paper positions at next mark). Released only by explicit `/kill release` with a typed confirmation phrase.

11.7 IDX microstructure guards [Z]

Order tickets are validated against live market rules before presentation: limit prices snap to valid tick fractions; the ticket warns when entry/stop sits within the day's auto-rejection bands; triggers never fire in the final 10 minutes of regular trading; illiquid names under the call-auction regime are excluded from the intraday engine entirely.

11.8 Crypto derivatives guards (Phase 4+, gated) [V]

If and only if a future compliance review approves derivatives: funding rate > +0.03%/8h with rising open interest deprioritizes new longs (long-squeeze flag); minimum liquidation buffer must exceed 3.5× daily ATR; execution friction modeled at ~0.02% maker / 0.05% taker plus depth-based slippage.

11.9 Safety checklist for every live proposal [P]

1. Instrument identity validated.
2. Price and account data are fresh.
3. Evidence timestamps do not leak future information.
4. Strategy version and risk policy are pinned.
5. Portfolio exposure after trade is within limits.
6. Stop/invalidation exists and is mechanically meaningful.
7. Worst-case notional and liquidity checks pass.
8. Venue rules and order precision pass.
9. No unresolved unknown orders/fills exist.
10. User approval is linked to the exact recommendation ID and constraints.

12. Trade Plan Schema & Human Template

12.1 Machine-readable contract [P][G][V, merged]

```
{
  "recommendation_id": "rec...",
  "instrument_id": "IDX:BBCA",
  "market": "IDX|CRYPTO",
  "decision": "WATCH|PAPER|LIVE_PROPOSAL|NO_TRADE",
  "direction": "LONG|SHORT|FLAT",
  "timeframe": "...",
  "thesis": "...",
  "entry": {"type": "price|condition", "value": "..."},
  "stop": {"price": "...", "why": "..."},
  "targets": [{"price": "...", "tier": 1}],
  "size": {"units": "lots|shares|contracts", "qty": 0, "notional": 0, "pct_account": 0, "amount_at_risk": 0},
  "risk_reward": 0.0,
  "invalidation": ["...", "...", "..."],
  "key_dates": ["..."],
  "weakest_assumption": "...",
  "chart_vs_research_disagreement": "...",
  "confidence": "LOW|MEDIUM|HIGH",
  "what_changes_confidence": "...",
  "numbers_to_verify": ["..."],
  "evidence_ids": ["ev..."],
  "risk_check_id": "rc...",
  "audit_hash_chain": {"prior_action_hash": "...", "observation_hash": "...", "action_hash": "..."},
  "generated_at": "2026-09-08T08:15:00+07:00",
  "expires_at": "...",
  "status": "AWAITING_HUMAN_REVIEW"
}
```

The optional `audit_hash_chain` block (SHA-256-linked, adopted from [V]) gives every plan a tamper-evident lineage without requiring blockchain infrastructure.

12.2 Human one-page template [G][Z]

```
TRADE PLAN — [TICKER] — [DATE]
THESIS · DIRECTION & TIMEFRAME · ENTRY · STOP (price + why) · TARGET 1 / 2
POSITION SIZE (lots/shares, IDR/USDT, % account, amount at risk) · RISK/REWARD
INVALIDATION (3 conditions) · KEY DATES · CHART vs RESEARCH DISAGREEMENT
CONFIDENCE (Low/Med/High) + what changes it · NUMBERS TO VERIFY BEFORE ACTING
APPROVE / REJECT (with optional note)
STATUS: AWAITING HUMAN REVIEW
```

13. Storage Model and API Contracts

13.1 Core Postgres tables [P][Z, merged]

Table	Purpose	Key fields
users	Owner account	id, auth_provider, preferences, timezone
portfolios	Paper/live portfolio	id, user_id, base_currency, mode, risk_policy_id
instruments	Ticker/coin registry	symbol, venue, asset_class, quote_currency, tick_fraction, lot_size
evidence	Provenance ledger	evidence_id, instrument_id, source, timestamps, payload_hash, quality
prices_daily / prices_intra	OHLCV	symbol, ts, ohlcv, source, adj_factor
fundamentals	Quarterly financials	symbol, period, revenue, earnings, DER, ROE, EPS, PBV, PER
news_items	Deduplicated corpus	url, published_at, title, source, tickers[], sentiment_hint
research_runs	Pipeline runs	run_id, workflow, status, started_at, model_versions, code_version
agent_artifacts	Per-agent outputs	run_id, agent_role, artifact_type, schema_version, payload
recommendations / plans	Trade plans	recommendation_id, run_id, status, thesis, invalidations, expiry
risk_checks	Risk gate results	recommendation_id, policy_version, checks, pass/fail, calculations
strategies	Strategy definitions	strategy_id, version, market, parameters, execution_model_id
backtest_runs	Backtest results	run_id, strategy_id, data_snapshot, metrics, leakage_verdict
paper_orders / live_orders	Simulated/real fills	order_id, portfolio_id, recommendation_id, status, fills
positions	Open positions	portfolio_id, instrument_id, qty, avg_price, unrealized_pnl
memory_decisions	Reflection log	symbol, date, verdict, pending, realized_return, alpha_return, lesson
journal_entries	Post-trade review	trade_id, thesis, outcome, deviation, lesson
alerts	Notifications	alert_id, severity, channel, status, acknowledged_at
audit_events	Immutable audit trail	event_id, actor, action, object_type, object_id, payload_hash
data_quality	Steward daily report	date, symbol, source_used, fallback_chain, bar_count, anomalies
heartbeats	Worker liveness	service, ts, version, kill_level

13.2 Tool/API boundaries [P]

Interface	Allowed	Forbidden
research.read	Fetch evidence and derived analytics	Placing orders
risk.check	Calculate and validate sizing/exposure	Changing its own policy

<code>portfolio.propose</code>	Create recommendation artifact	Sending exchange order
<code>execution.precheck</code>	Validate an explicit order request against hard limits	Interpreting free text
<code>execution.submit</code>	Submit an approved structured order	Taking an LLM-generated sentence as order input
<code>execution.reconcile</code>	Read order/fill/account state	Silently retrying unknown orders
<code>journal.write</code>	Record outcome and review	Editing raw execution state

All timestamps stored in UTC with Asia/Jakarta offset applied only at presentation boundaries, to eliminate dual-timezone bugs across WIB sessions and 24/7 crypto. [Z]

14. Deployment Architecture

14.1 Recommended topology [Z][P]

Laptop-first (or VPS/Docker) for the worker + Hermes gateway; Supabase as the cloud state/control plane; Firebase optional, client/event layer only. This follows directly from a verified constraint: neither Supabase Edge Functions (150–400s wall-clock) nor Firebase Cloud Functions (540–3,600s) can host a long-running Python agent process.

14.2 Decision matrix: Supabase vs Firebase [Z][P]

Criterion	Supabase	Firebase	Winner
Relational trading data	Excellent	Good, but document-centric	Supabase
RLS / row authorization	Excellent paradigm	—	Supabase
Realtime UI	Strong	Excellent	Tie
Scheduled jobs	Strong (pg_cron/pg_net)	Strong (Cloud Scheduler)	Tie
Quant/Python ecosystem	Easy via external workers	Usually external workers	Supabase
Vector/RAG memory	pgvector native	External service	Supabase
Fast mobile-first UX	Good	Excellent	Firebase
Recommendation	PRIMARY	Optional client/event layer	Supabase

14.3 Laptop worker (Docker Compose) [Z]

Single `docker-compose.yml` with four services: `mbg-worker` (Python app), `hermes` (agent gateway), `redis` (debounce/queue), and no inbound ports required (Telegram uses long polling; Supabase is outbound-only). Restart policy `unless-stopped` on every service; the worker writes a heartbeat row every 60s and a watchdog restarts any pipeline node silent longer than its checkpoint timeout.

14.4 MBG Lite — zero-infrastructure onboarding tier [Z]

A single Hermes skill bundle — installable with one command into Hermes, OpenClaw, or Claude Code — containing the five-step Astra workflow as SKILL.md prompts with the safety rules (§3) baked in. The user pastes dated data per ticker; the skill enforces FACTS/OPINION separation, the `NOT PROVIDED` rule, confidence labels, the verify-list, and the approve/reject ritual, saving each finished plan as a markdown file that later seeds the full system’s backtester. **This exists so the product is useful on day zero, before the worker stack is deployed, and so the user builds the verification habit before automation removes the friction.**

14.5 Secrets and key hygiene [Z]

LLM API keys live only in the laptop `.env`, injected into containers at runtime. The Supabase service key never appears on the dashboard (anon key + RLS bound to the owner’s auth UID only). Exchange keys, when Phase 3+ arrives, use trading-disabled-then-enabled scope progression with IP restrictions where supported. No secret is ever logged — a trace-redaction layer strips key-shaped strings from all bot transcripts before they reach the dashboard.

15. UX, Reports, and User Commands

15.1 Daily recommendation packet [P]

```
MBG DAILY DESK - YYYY-MM-DD
1. REGIME          IDX: ...   CRYPTO: ...
2. WHAT CHANGED    - ...
3. TOP CANDIDATES  1) ... 2) ...
4. RESEARCH VERDICTS  Bull vs Bear disagreement: ...
5. RISK MAP         Exposure / concentration / event risk / liquidity
6. ACTIONABLE PLANS  WATCH / PAPER / LIVE_PROPOSAL
7. NO-TRADE CONDITIONS ...
8. DATA QUALITY    PASS / WARN / BLOCK
9. VERIFY BEFORE ACTION ...
```

The brief must be generated even when some agents fail — degraded sections are labeled as such, never silently omitted, because a silently-missing section reads as “nothing happening.” [Z]

15.2 User commands (Telegram / CLI) [Z][P]

Command	Behavior
<code>/brief</code>	Fetch latest daily desk output (both markets)
<code>/scan IDX / /scan CRYPTO</code>	Scan the configured universe
<code>/analyze TICKER</code>	Queue a deep-analysis run; director-style NL accepted (“bank stocks under 5000”)
<code>/plan TICKER</code>	Latest plan card
<code>/backtest STRATEGY RANGE</code>	Run reproducible backtest and report
<code>/portfolio</code>	Positions, equity, open risk
<code>/shadow</code>	Compare decisions with observed account behavior
<code>/risk</code>	Portfolio risk and active breaches
<code>/watch ADD REMOVE</code>	Manage watchlist
<code>/approve PLAN_ID / /reject PLAN_ID</code>	Sign-off with optional note
<code>/status</code>	Worker heartbeat, data freshness, model/cost meters
<code>/kill</code>	Arm or release the system-wide trading kill-switch

15.3 Dashboard views [P][Z]

Daily Desk (regime, candidates, evidence quality, open proposals) · Portfolio (exposure, drawdown, risk budget, correlation, positions/orders) · Research Run (agent tree, evidence, debate transcript, risk map — the transparency feature) · Backtest Lab (strategy versions, metrics, walk-forward, stress tests, reproducibility hash) · Journal (planned vs actual, slippage, adherence, lessons) · System Health (providers, queues, model usage, failed gates, stale data, incident log).

16. Implementation Roadmap

Merged from the three phased roadmaps (Grok, GLM, GPT/v5) into one sequence, preserving GLM’s week-scale granularity and exit criteria while folding in Grok’s credit-assignment phase and the optional v5 econometric module. **The sequence enforces one trust arc: data before analysis, analysis before advice, advice before measurement, measurement before automation.**

Phase	Weeks	Focus	Deliverables	Exit criteria
P0 — Foundation	1-2	Hermes env + safety + data skeleton	Compose stack, Postgres schema, <code>mbg/</code> skill folder with safety rules in every prompt, one IDX + one crypto data adapter, heartbeat + kill-switch skeleton, MBG Lite bundle published	10-year <code>.JK</code> history for 30 tickers cached with zero gaps; heartbeat visible in Supabase
P1 — Research Desk / Daily Brief MVP	3-4	One-shot + core debate	<code>mbg-research-desk</code> , <code>debate-orchestrator</code> , adversarial bull/bear, <code>research-manager</code> , <code>risk-manager</code> , <code>/brief</code> <code>/analyze</code> <code>/status</code>	Brief delivered 5 consecutive trading days by 07:05 WIB, all sections present
P2 — Gating, Regime & Backtest/Paper	5-8	Quality filters + advice core	<code>conviction-gater</code> , <code>regime-detector</code> , full risk committee, plan cards, paper broker with T+2, vectorbt replay, walk-forward + stress-test suite	20 plans paper-tracked over 10 sessions with correct fills/costs; guard rejection path proven

P3 — Daily Pipeline, Memory & Credit	9–10	Continuous operation + learning	Scheduled orchestrator, strategist-memory reflection loop, dashboard with Realtime, shapley-credit-assigner , Opportunity Scanner (F-10), event-timing risk gate	First lessons injected into PM prompts and visible in transcripts; Shapley weights computed on ≥1 batch of closed trades
P4 — Hardening & Guarded Live	11–14	Backtest depth + deploy + execution	backtest-lab (Monte Carlo, evolution backtest), Docker/Supabase/Firebase adapters finalized, deterministic execution gateway, idempotency + reconciliation, one approved broker/venue adapter, small-notional manual-approval live mode	30-day reliability report ≥ targets; incident + kill-switch drills passed; ≥60 scored paper plans across varied conditions before automation work begins
P5 — Scale / Intelligence (optional)	15+	Portfolio intelligence	Portfolio optimizer, cross-market allocation, regime ensemble, bounded strategy evolution, econometric proofer (DSR/PBO/Hansen SPA) , derivatives compliance review (if pursued), multi-venue execution	Only after stable observation period + documented rollback/runbook

Definition of Done for MVP (P0–P2): a single command produces a full, sourced, auditable recommendation packet for IDX and crypto; the same strategy definition can be backtested with reproducible results and leakage checks; every proposal is sized by the deterministic risk service and can be vetoed; recommendations become simulated orders with fills and journal entries; Hermes skills can discover/run workflows with concise operator commands; health, logs, kill-switch, and incident workflow are tested. [P]

17. Success Metrics

17.1 KPIs [Z, extended]

KPI	Definition	Target
Brief punctuality	% trading days brief lands ≤ 07:05 WIB	≥ 95%
Pipeline success	% runs completing (resume included)	≥ 98%
Plan quality (paper)	% scored plans with positive alpha vs ^JKSE at T+5	≥ 45% (honest floor with positive expectancy at 1.5:1+ R:R — not a coin-flip target)
Guard integrity	Plans published that violated the rulebook	0
Data integrity	Mornings with unresolved steward anomalies	≤ 1/month
Cost discipline	Mean daily LLM spend	≤ \$0.35/day
Learning signal	Memory lessons cited in later PM decisions	≥ 60% of decisions
Credit stability	Shapley weights stable after burn-in; dummy agents near zero	tracked, no hard target pre-P3
Safety (absolute)	Unauthorized live orders	0 , always

17.2 Acceptance scenarios (executed verbatim at each phase exit review) [Z]

- **AS-1 Morning brief** — worker online, 20 tickers configured; 07:00 WIB cron fires; Telegram receives full brief by 07:05, every section present or explicitly degraded, HTML archive resolves, spot-checked against **analyses** table on 3 tickers.
- **AS-2 Plan lifecycle** — BBCA analysis produces a Buy verdict; plan card publishes with entry/stop/target/size; Risk Guard validation passes; “Simulate fill” books a paper order at next open with correct fees/levy; position appears on dashboard within one Realtime tick; T+2 ledger blocks same-day reuse of proceeds.
- **AS-3 Memory loop** — a scored plan whose verdict was wrong (alpha < 0 at T+5) triggers a 2–4 sentence lesson from the steward; the next analysis of the same ticker cites it in **past_context**; dashboard transcript shows the citation verbatim.

17.3 Monthly operating cost budget [Z]

Item	Assumption	Monthly cost
LLM API	20 tickers/day deep pipeline + debates + memory reflections, fast tier + deep judges	\$2–6
Supabase	Free tier (500MB fits ~2 years of daily rows)	\$0

Dashboard hosting	Vercel hobby tier	\$0
Telegram	Bot API	\$0
Market data	yfinance + RSS + CoinGecko demo + ccxt public	\$0
Electricity/laptop	Always-on mini workload (~15W avg)	~\$1-2
Total (recommended config)	fast tier + deep-tier judges only	< \$10/month

18. Open Decisions & Resolved Conflicts

These are the points where the four source drafts genuinely disagreed. Each is called out explicitly rather than silently averaged, so you can override any of them before implementation begins.

18.1 Runtime shape: “Hermes does everything” vs “Hermes orchestrates, Python computes”

Resolved: deterministic services behind Hermes skills (§4.1), following [P]’s reasoning that LLM output must never be the arithmetic itself. [G] and [Z]’s concrete SKILL.md/bot-profile designs are preserved as the *implementation* of that orchestration layer, not overridden.

18.2 Crypto scope: spot-only (OJK-whitelisted) vs perpetuals on Hyperliquid/OKX

Resolved in favor of spot-only for v1. [Z] and [P] independently researched and cited the actual OJK licensing regime (POJK 27/2024, POJK 23/2025, PADK 3/2026) and its whitelist; Binance and most perp venues are not licensed for Indonesian residents. [V]’s perpetuals-first design (funding rate drag, liquidation clusters, cross-margin) is a strong analytical module but is **retained only as a Phase 5+, compliance-gated feature** — the analytical bot (`mbg-crypto-specialist`) and schemas are kept, but derivatives fields stay dormant until a dedicated legal/compliance review approves a specific venue.

18.3 Naming: “Market Brain Grid” vs “Market Brain Guardian”

[P]’s draft proposed renaming the product to “Market Brain Guardian” for brand clarity. **Rejected** — three of four drafts and the original brief use “Market Brain Grid (MBG)”; kept as-is. If a rename is wanted later, “Market Brain Guardian” is on record as the top alternative.

18.4 Autonomy posture at the math-gate layer

[V]’s “Unanimous Gatekeeper Approval” (DSR/RiskGuard veto) reads as *sufficient* for execution once math passes. **Resolved:** econometric/risk gates are additional vetoes, not substitutes for human approval — every live proposal still requires `/approve PLAN_ID` regardless of how clean its DSR/PBO numbers are. This preserves the non-goal stated in every draft: no unsupervised live capital deployment.

18.5 Minimum R:R threshold

[G]/[Z] use “under ~2:1 is not worth it”; [Z]’s risk table separately states “Min RR ratio (swing) = 1.5”. **Resolved:** keep both as asset/context-dependent — 1.5:1 floor for swing/position trades, but the plan must explicitly flag anything under 2:1 as marginal rather than silently passing it. This is now stated as such in §3 rule 6 and §11.2.

18.6 Econometric proofing module: mandatory or optional?

[V] treats DSR/PBO/Hansen SPA as core (Stage 3 of its 5-stage pipeline). Given this is a single-strategy v1 launch, treating it as mandatory before any trade exists is premature — there’s no strategy library yet to overfit across. **Resolved:** optional, off by default, activated in Phase 5+ once there’s enough strategy/trial history for “probability of backtesting overfitting” to be a meaningful number.

18.7 Items still requiring a human decision before coding the execution layer [P]

1. First live venue for crypto (Indodax vs Tokocrypto vs later addition) and first IDX broker/data adapter.
2. Default risk budget and portfolio limits, if different from the §11.2 defaults.
3. LLM provider(s) and per-role model tiers.
4. Retention/privacy policy for research and account data.
5. Confirm local-first vs cloud-first for day-one usage.
6. Exact live-mode approval ceremony and emergency policy wording.
7. Whether to pursue the crypto-derivatives compliance review at all (§18.2) or keep it permanently out of scope.
8. IDR stablecoin quote convention for crypto paper P&L — USDT via Indodax, or direct IDR pairs. **[Z]**

19. Risk Register (merged)

Risk	Likelihood	Impact	Mitigation	Source
LLM hallucination / invented numbers	Med	High	Universal NOT PROVIDED rule; tool-first data; FACTS vs OPINION; verified-snapshot grounding tool; human verify	[G][Z][P]
Look-ahead / survivorship bias in backtests	Med	High	Point-in-time data layer; anti-leakage checklist; automated look-ahead sentinel tests	[P][Z]
yfinance/idx.co.id endpoint changes or throttling	Med	High	Ordered fallback chain; bar-count alarm; batched polling with backoff; curl_cffi impersonation + golden fixtures	[Z]
Debate cost / latency	Med	Med	Cap rounds; cheap models for analysts; mode selection by need	[G][Z]
Cold-start Shapley credit assignment	High (early)	Low	Bayesian mixture with uniform prior until sample size sufficient	[G]
Over-trading / revenge trading	Low	High	Conviction gating, daily loss breaker, human gate, kill-switch	[G][Z][V]
Regulatory shift (OJK rules, IDX band changes)	Med	Med	Rules stored as versioned config data, not hard-coded; quarterly rules-review task on roadmap	[Z][P]
Laptop downtime misses sessions	High	Low	Cloud EOD fallback snapshot; catch-up replay on reconnect; missed sessions scored as no-trade	[Z]
Single-builder bus factor	Med	Med	Everything in git; skills portable to OpenClaw/MOSS/Claude Code; this PRD is the recovery manual	[Z]
Overfitting the daily loop to a recent regime	Med	Med	Walk-forward gate before config changes; mandatory benchmark; memory lessons tagged with regime context	[P]
Crypto derivatives compliance exposure	Low (v1, out of scope)	High if activated	Feature dormant until dedicated legal review (§18.2)	[Z][P] vs [V]

20. References

User-supplied source PRDs (this document's lineage): - Grok v1.2 — PRD_Market_Brain_Grid_MBG_v1_2_Grok.pdf - GPT/ChatGPT Final — Market_Brain_Grid_PRD_Final_GPT.pdf - GLM v1.1 (Z.ai) — Market_Brain_Grid_PRD_v1_1_GLM_.pdf - Gemini v5.0 — Market_Brain_Grid_PRD_v5_Gemini.pdf - Original ChatGPT Astra AI Trading Bot guide (referenced by all four drafts) — user-supplied attachment; SCAN→ANALYZE→RESEARCH→RISK→PLAN, human review, no automatic execution.

Reference projects and papers (as cited across the four drafts): - Xiao, Sun, Luo, Wang — *TradingAgents: Multi-Agents LLM Financial Trading Framework*, arXiv:2412.20138 - TauricResearch/TradingAgents — github.com/tauricresearch/tradingagents (Apache-2.0) - The-Swarm-Corporation/AutoHedge — github.com/The-Swarm-Corporation/AutoHedge (MIT) - HKUDS/Vibe-Trading — github.com/HKUDS/Vibe-Trading, vibetrading.wiki (MIT) - moss-site/moss-trade-bot-skills — github.com/moss-site/moss-trade-bot-skills (MIT-0), moss.site - NousResearch/hermes-agent — github.com/NousResearch/hermes-agent (MIT) - CLQT closed-loop quant trading standard — arXiv:2606.29771 (as cited in the v5 draft) - Bailey & López de Prado — Deflated Sharpe Ratio, CSCV/PBO, Hansen SPA methodology - OJK — POJK 27/2024, POJK 23/2025 (amendment), PADK 3/2026 (implementation), transition-completion notice - PMK 68/2022 — Indonesian income tax and VAT treatment of crypto asset trades - IDX (idx.co.id) — trading rules: sessions, tick fractions, auto-rejection bands, settlement - Supabase and Firebase — platform quota/pricing documentation

Market Brain Grid (MBG) — Unified PRD v2.0 — End of document. This is the single source of truth for implementation. Start with Phase 0-1 (safety + research desk + adversarial debate + MBG Lite). Every output remains research-grade until a human changes STATUS from AWAITING HUMAN REVIEW. Not financial advice. Educational and research purposes only. Verify every number. Investing involves risk of loss.